

FAHD KHATTAK

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Education

Louisiana State University | Roger Hadfield Ogden Honors College

May 2027

B.S. Computer Science | Business Administration Minor

Baton Rouge, LA

Relevant Coursework: Operating Systems, Object Oriented Design, AI & LLMs, Data Structures, Numerical Methods

Awards & Honors: LSU President's Alumni Scholar, President's List ('23-'25), LSU President's Future Leaders in Research

Skills

Languages: Python, Verse, Bash, SQL, C, C++, Java, HTML, CSS, JavaScript, TypeScript, YAML

Technology/Frameworks: AWS, Azure, Kubernetes, Node.js, React, ChromaDB, PostgreSQL, NumPy, TensorFlow, Unsloth, Pandas, QT, Git, Jest, Pytest, Jira, Agile, Excel, CUDA, Linux, Unreal Engine 5, Unity

Experience

JP Morgan Chase

June 2026 – Aug 2026

Software Engineer Intern

Houston, Texas

- Incoming Software Engineer Intern
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Northrop Grumman

June 2025 – Aug 2025

Software Engineer Intern

Dulles, Virginia

- Drove a **430% increase** in max simulation runtime supporting **14+ concurrent headless scenarios** with multi-instructor access by replacing legacy Windows VMs with a **PostgreSQL-powered client-server architecture** on a **Kubernetes cluster**
- Streamlined deployments and cut integration errors for **6 downstream teams** working with a multiple-TB database by introducing a **Python-driven Alembic schema versioning system** across a **PostgreSQL stack**
- Collaborated in a fast-paced **Agile** team environment, participating in sprint planning and code reviews while delivering iterative improvements to simulation runtime, database reliability, and GUI tools

Shell USA

Jan 2025 – May 2025

Machine Learning Intern

Baton Rouge, Louisiana

- Launched real-time analytics of railcar lock conditions via a centralized **Power BI dashboard** continually fed with model outputs through an end-to-end pipeline
- Delivered **85% classification accuracy** on defective railcar locks by fine-tuning **Llama-3.1-Vision-Instruct** with **Python, QLoRA, Unsloth, and Hugging Face**
- Achieved **sub-10s inference** and reduced manual inspections by dockerizing a **4-bit quantized vision model**, exposing it via a **REST API**, and integrating **RAG-based retrieval from Azure Blob Storage**

SmartSense (LSU Biomechanics Lab)

Aug 2024 – Present

Backend Software Engineer

Baton Rouge, Louisiana

- Enabled **OSHA-compliant reporting** with **<10ms latency** for **10+ endpoints** by building a real-time safety platform using **React.js** and **Node.js**
- Streamlined backend workflows with **JavaScript** for real-time data logging and multi-region payment processing by building **AWS Lambda** functions integrating **DynamoDB** and the **Stripe API**
- Improved deployment reliability with **100% branch coverage** by writing comprehensive **Jest** unit tests across the JavaScript codebase

Projects

Epic Games

June 2023 – Present

Game Developer

Remote

- Collaborated on a **5-person studio** to create Fortnite maps in **Unreal Engine 5** using **C++/Verse** and **object-oriented design**
- Reached **10M+ total plays**, **10k peak CCU**, and generated **\$250k in revenue**
- Sustained **2,000+ concurrent users** and boosted retention by **30%** through rapid, bug-free iterations improving UI/UX, gameplay systems, and animations

Leadership

Engineering Development Ambassadors (EDA)

May 2023 – Present

Outreach Chair, Treasurer

- Mentored **200+** club members by leading engineering demos, activities, and discipline talks at **K-12 schools**
- Managed a **\$30K/year budget**, ensuring smooth and compliant financial operations